15. A wheat variety, Atlas 66 which has been used as a donor for improving cultivated wheat, which is rich in
    a) iron b) carbohydrates
    c) proteins d) vitamins
16. Which one of the following crop varieties correct matches with its resistance to a disease?

Variety	Resistance to disease
a) Pusa Komal	Bacterial blight
b) Pusa Sadabahar	White rust
c) Pusa Shubhra	Chilli mosaic virus
d) Brassica	Pusa swarnim

17. Which of the following is incorrectly paired?
    a) Wheat - Himgiri
    b) Milch breed - Sahiwal
    c) Rice - Ratna
    d) Pusa Komal - Brassica
18. Match list I with list II

List I	List II
Biofertilizer	**Organisms**
i) Free living N2	a) *Aspergillus*
ii) Symbiotic N2	b) *Amanita*
iii) P Solubilizing	c) *Anabaena azollae*
iv) P Mobilizing	d) *Azotobactor*

    a. ic, iia, iiib, ivd b. id, iic, iiia, ivb.
    c. ia, iic, iiib, ivd c. ib, iia, iiid, ivc.
19. Differentiate primary introduction from secondary introduction.
20. How are microbial innoculants used to increase the soil fertility?
21. What are the different types of hybridization?
22. Explain the best suited type followed by plant breeders at present?
23. Write a note on heterosis.
24. List out the new breeding techniques involved in developing new traits in plant breeding.

Glossary

Acclimatization : The adaptation of an individual to a changed climate or the adjustment of a species or a population to a changed environment over a number of generations.

Agronomy : Science of farming

Germplasm Collection : The entire collection (of plants / seeds) having all the diverse alleles for all genes in a given crop is called **germplasm collection.**

Non recurrent parent : The parent of a hybrid that is not again used as a parent in backcrossing

Quarantine : Strict isolation imposed to prevent the spread of disease

Strain : A group of similar individuals from a common origin.

APPENDIX

Vavilov's Centre of Crop Origin		Crops domesticated
1	China	Foxtail millet, soybean, bamboo, onion, crucifers.
2	India	Rice, sugarcane, mango, orange, eggplant, sesame.
2 a	South East Asia	Rice, banana, coconut, clove , hemp.
3	Central East	Wheat, pea, hemp, cotton etc.
4	The Near East	Wheat, rye, many subtropical and tropical fruits.
5	Mediterranean	Olive, vegetables, oil yielding plants, wheats
6	Ethiopia (Abyssinian)	Wheat, barley, sesame, castor, coffee.
7	Mesoamerica (South Mexican & Central American Centre)	Maize, bean, sweet potato, papaya, guava, tobacco.
8	South America	Tomato, pine-apple
8 a	The Chiloe Centre	Potato
8 b	The Brazilian –Paraguayan Centre	Groundnut, cashew nut, pine apple, peppers, rubber.

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